AI-use disclosure — Quantum Traffic Gridlock (team: Arif, Rohit, Mehul)

Problem statement section 10 permits AI assistants provided assistance is declared and results are independently verified. This page declares all AI assistance in this submission. The detailed, dated record is `AI_USE_LOG.md`.

1. Tools and what each did

tool	what it did
Claude (Claude app)	Explained the problem statement and the quantum concepts to the team (all three of us are beginners in quantum computing); built the numpy prototype in `research/` that compared QUBO surrogates and QAOA training variants; wrote `traffic_qubo.py` (queue recursion, causal windows, local quadratic projection, QUBO and Ising conversion) and the project brief `CLAUDE.md`; reviewed our `metrics.json` with us and diagnosed the wrong ramp sign (section 3); typesets the PDF versions of the report and this disclosure.
Claude Code (model: Claude Sonnet 5.5)	Wrote the tests (`tests/`), `qaoa.py`, `baselines.py`, `run.py`, `scenario_gen.py`, `ablations.py`, `generalisation.py`, `report_numbers.py` and `make_outputs.sh`; ran them in our repository; drafted `README.md` and `REPORT.md` from the files in `outputs/` and `logs/`.

No other AI tool was used. Libraries used (permitted): Qiskit, NumPy, SciPy.

2. What the team did

- Reviewed and approved the design (QUBO rule, QAOA depth, objective, optimiser, resource limits) and **approved the ablation selection rule before the ablation was run** (change the ramp scale only if mean lift on 30 random scenarios rises by ≥ 0.02 without lowering the minimum lift; R and H are never used to choose settings).
- Ran every test and experiment on our own machine and looked at the outputs.
- Checked the generated schedules with the official `scorer.py`.
- Followed the organisers' Discord announcements about scenario H (released late, about 17:40 IST on 16 Oct), reviewed the sign-correction evidence with Claude and **approved the change** (see section 3).
- Reviewed the report against the logs before submission.

3. Late sign error in the QAOA ramp (found by AI review, fixed with team approval)

Claude (Claude app) noticed initial_objective = +1.45 in metrics.json during review and diagnosed the ramp sign; Claude Code verified it with an ablation and implemented the fix; the team approved the change. The change was judged by the same selection rule fixed in advance (30 random scenarios only); the earlier wrong-sign results were kept in the ablation logs as a separate "before the sign fix" block, and the depth, random-angle, surrogate and CVaR ablations were rerun on the corrected ramp.

4. How results were verified independently of the AI-written code

- **Official scorer:** the four reference vectors (116.0, 146.0, 237.0, 92.5) are a unit test; every cost in the report is computed by `official/scorer.py`; `scorer.py --schedule outputs/R/seed0/schedule.json` runs on our `schedule.json`.
- **Model check:** our queue recursion equals the scorer on 7,936 evaluations (R and 30 random scenarios × 256 schedules, 0 mismatches, `logs/verification.json`); the QUBO equals the global least-squares quadratic fit of the official cost (maximum difference 1.9e-12).
- **Tests (20, about 4 s):** bit mapping round trip, qubit order in a real circuit, `qubo.json` energies, counts sum to 4,096 with 8-bit keys, circuit statevector vs counts (total variation within 1.5× pure shot noise), `circuit.qasm` equals the dump of the circuit rebuilt from the saved angles, E_Q / lift / enrichment recomputed from `counts.json` and the scorer, ≤ 300 training evaluations, and the initial QAOA angles must start below the random energy level (added after the late sign error, see section 3).
- **Determinism:** the same seed gives byte-identical `counts.json` and `circuit.qasm` (tested); re-running `make_outputs.sh` changes only the `runtime_seconds` field of `metrics.json`.
- **Report numbers** are produced by scripts into `outputs/` and `logs/`, not typed from memory.

5. Integrity statements

- The QUBO comes from a documented, scenario-independent traffic rule (local terms over causal windows, best quadratic approximation per term, exact switching term). It was **never** built from enumeration, from the optimal schedule or from any scenario-specific lookup.
- Enumeration of the 256 schedules is used only for benchmarks (C*, C_rand, elite set, regret) after the fact; it never feeds the QUBO or the QAOA training.
- Nothing is hard-coded for scenario R. R, H and our own held-out scenarios are not used to choose settings (the ramp scale was chosen on 30 random scenarios).
- Classical refinement appears only as a separately labelled "hybrid" result, never in the raw QAOA counts.
- Files in `official/` are unmodified. Scenario H was run unmodified after all settings were fixed; no setting was chosen on R or H.
- Known weaknesses (a wrong-sign mixer angle shipped until a late review and is now fixed, training hits the 300-evaluation cap, surrogate regret up to 32.6 % on some scenarios, selection noise in the ramp choice) are stated in `REPORT.md` section 7.